The James-Stein Estimator and Stein's Phenomenon

STAT 220B – Lecture 10

Where we left off

Lecture 9 built the machinery:

- **Setup.** $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$, squared-error loss. $\delta^0 = \mathbf{x}$ has constant risk p.
- **Stein's Identity.** $\mathbb{E}_\theta[(\mathbf{X} - \theta)^T \mathbf{g}(\mathbf{X})] = \mathbb{E}_\theta[\nabla \cdot \mathbf{g}(\mathbf{X})]$.
- **SURE.** For $\delta = \mathbf{x} + \mathbf{g}(\mathbf{x})$: $\hat{R}(\mathbf{x}) = p + 2\nabla \cdot \mathbf{g}(\mathbf{x}) + \|\mathbf{g}(\mathbf{x})\|^2$ unbiasedly estimates the risk.
- **For the James-Stein estimator** $\delta^{JS}(\mathbf{x}) = (1 - (p-2)/\|\mathbf{x}\|^2)\mathbf{x}$ with $\mathbf{g}(\mathbf{x}) = -(p-2)\mathbf{x}/\|\mathbf{x}\|^2$:

$$\hat{R}^{JS}(\mathbf{x}) \;=\; p \;-\; \frac{(p-2)^2}{\|\mathbf{x}\|^2}.$$

Today: complete the proof, examine the risk gain, build the positive-part improvement, and discuss what Stein's phenomenon means for statistical practice.

Reading: Berger, Section 5.4.3 (multivariate normal mean estimation).

Proving the James-Stein dominance

What remains

Yesterday's SURE computation gave

$$R(\theta, \delta^{JS}) \;=\; \mathbb{E}_\theta[\hat{R}^{JS}(\mathbf{X})] \;=\; p \;-\; (p-2)^2\, \mathbb{E}_\theta\left[\frac{1}{\|\mathbf{X}\|^2}\right].$$

To conclude $R(\theta, \delta^{JS}) < R(\theta, \delta^0) = p$, we need:

$$\mathbb{E}_\theta\left[\frac{1}{\|\mathbf{X}\|^2}\right] \;>\; 0 \quad \text{for all } \theta \in \mathbb{R}^p, \;\text{ and finite when } p \geq 3.$$

The positivity is automatic since $1/\|\mathbf{x}\|^2 > 0$. The substantive content is finiteness and the explicit form.

Distribution of $\|\mathbf{X}\|^2$

For $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$, the squared norm $\|\mathbf{X}\|^2$ follows a **noncentral chi-squared distribution** with p degrees of freedom and noncentrality parameter $\lambda = \|\theta\|^2$:

$$\|\mathbf{X}\|^2 \sim \chi^2_p(\lambda), \qquad \lambda = \|\theta\|^2.$$

Mixture representation. A noncentral chi-squared admits a Poisson mixture of central chi-squareds:

$$\|\mathbf{X}\|^2 \stackrel{d}{=} \chi^2_{p+2K}(\text{central}), \quad \text{where } K \sim \text{Poisson}(\lambda/2).$$

(Conditioning on $K = k$, $\|\mathbf{X}\|^2$ has a central chi-squared distribution with $p + 2k$ degrees of freedom.)

Evaluating the inverse-norm expectation

Using the Poisson mixture and the moment formula $\mathbb{E}[1/\chi^2_\nu] = 1/(\nu - 2)$ for $\nu > 2$:

$$\mathbb{E}_\theta\left[\frac{1}{\|\mathbf{X}\|^2}\right] = \mathbb{E}\left[\frac{1}{\chi^2_{p+2K}}\right] = \mathbb{E}_K\left[\frac{1}{p-2+2K}\right],$$

where $K \sim \text{Poisson}(\|\theta\|^2/2)$.

Finiteness for $p \geq 3$. The denominator $p - 2 + 2K \geq p - 2 > 0$ always (since $K \geq 0$), so the expectation is bounded by $1/(p-2) < \infty$. The expectation is also strictly positive.

Closed form. For $p \geq 3$:

$$\mathbb{E}_\theta\left[\frac{1}{\|\mathbf{X}\|^2}\right] = e^{-\|\theta\|^2/2} \sum_{k=0}^{\infty} \frac{(\|\theta\|^2/2)^k}{k!\,(p-2+2k)}.$$

The risk formula for δ^{JS}

James-Stein Risk Theorem

For $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$ with $p \geq 3$, under squared-error loss:

$$R(\theta, \delta^{JS}) \;=\; p \;-\; (p-2)^2\, e^{-\|\theta\|^2/2} \sum_{k=0}^{\infty} \frac{(\|\theta\|^2/2)^k}{k!\,(p-2+2k)}.$$

In particular, $R(\theta, \delta^{JS}) < p = R(\theta, \delta^0)$ for every $\theta \in \mathbb{R}^p$.

Corollary. δ^0 is **inadmissible** for $p \geq 3$.

This is the Stein phenomenon (Stein 1956; James and Stein 1961). The sample mean, the canonical "obvious" estimator of a multivariate normal mean, is uniformly improvable in three or more dimensions.

Two extreme cases

Case 1: $\theta = 0$ (origin). Then $K = 0$ a.s. (Poisson with mean 0), so

$$R(0, \delta^{JS}) \;=\; p \;-\; \frac{(p-2)^2}{p-2} \;=\; p \;-\; (p-2) \;=\; 2.$$

The risk gain at $\theta = 0$ is $p - 2$, which is enormous when p is large. For $p = 1000$ unrelated normal means, shrinkage near the origin reduces risk from 1000 to 2.

Case 2: $\|\theta\| \to \infty$. Then $K \to \infty$ in probability, $(p - 2 + 2K)^{-1} \to 0$, and

$$R(\theta, \delta^{JS}) \;\to\; p.$$

The gain vanishes far from the origin. There is no free lunch globally: δ^{JS} improves on δ^0 everywhere, but the improvement concentrates near the shrinkage target.

Numerical comparison

For $p = 10$, $\|\theta\|^2 = \lambda$:

$\|\theta\|^2$	$R(\theta, \delta^{JS})$	Gain over δ^0
0	2.000	80.0%
5	4.78	52.2%
10	6.21	37.9%
50	8.86	11.4%
100	9.40	6.0%
1000	≈ 10	$\approx 0\%$

The improvement is *uniform* but its *magnitude* depends on $\|\theta\|^2$. Practitioners often interpret this as: shrinkage helps most when you guess the center well.

Geometry of the Stein phenomenon

The thin-shell intuition

Why does shrinkage help in dimension ≥ 3?

For $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$:

$$\mathbb{E}_\theta \|\mathbf{X} - \theta\|^2 \;=\; p, \qquad \mathrm{Var}_\theta(\|\mathbf{X} - \theta\|^2) \;=\; 2p.$$

So $\|\mathbf{X} - \theta\| \approx \sqrt{p}$ with fluctuation of order 1. The distribution concentrates on a **thin spherical shell** of radius $\sqrt{p}$ around θ.

Consequence: $\mathbf{X}$ is almost never close to θ. It is almost always at distance $\sqrt{p}$.

This is counterintuitive in low dimensions but is a basic feature of high-dimensional Gaussians. The volume of $\mathcal{N}_p(\theta, I_p)$ near θ itself is negligible compared to the volume in the shell.

Why shrinking helps

If $\mathbf{X}$ is on a shell at distance $\sqrt{p}$ from θ, then $\|\mathbf{X}\|$ is typically larger than $\|\theta\|$ (the shell extends outward as well as inward). Shrinking $\mathbf{X}$ toward $\mathbf{0}$ moves the estimate inward, partially correcting this overshoot.

Quantitatively: if $\theta = 0$, then $\mathbf{X}$ is at distance $\sqrt{p}$ from the origin, while the truth is at the origin. Multiplying by $(1 - c/\|\mathbf{X}\|^2)$ for appropriate c shrinks toward 0 and reduces the error.

The Stein factor $c = p - 2$ is precisely tuned: smaller and we shrink too little; larger and we overshoot through zero.

A risk decomposition

The mathematics of risk admits a bias-variance decomposition for $\delta^c(\mathbf{x}) = (1 - c/\|\mathbf{x}\|^2)\mathbf{x}$:

$$R(\theta, \delta^c) \;=\; p \;-\; c(2(p-2) - c)\,\mathbb{E}_\theta\left[\frac{1}{\|\mathbf{X}\|^2}\right].$$

(One can verify this with SURE: $\mathbf{g}(\mathbf{x}) = -c\mathbf{x}/\|\mathbf{x}\|^2$, $\nabla \cdot \mathbf{g} = -c(p-2)/\|\mathbf{x}\|^2$, $\|\mathbf{g}\|^2 = c^2/\|\mathbf{x}\|^2$.)

Minimizing the right side over c gives $c^* = p - 2$. Hence $c = p - 2$ is the **optimal** shrinkage factor among the linear-shrinkage family $(1 - c/\|\mathbf{x}\|^2)\mathbf{x}$.

So δ^{JS} is not arbitrary: it is the optimum within its parametric class.

An obvious improvement: the positive-part Stein estimator

A defect of δ^{JS}

When $\|\mathbf{x}\|^2 < p - 2$, the shrinkage factor $1 - (p-2)/\|\mathbf{x}\|^2$ is negative. The James-Stein estimator then *reverses the sign* of $\mathbf{x}$.

Concrete example. Suppose $\mathbf{x} = (0.1, 0.1, 0.1)$ in $\mathbb{R}^3$. Then $\|\mathbf{x}\|^2 = 0.03$, so the shrinkage factor is $1 - 1/0.03 \approx -32$, and

$$\delta^{JS}(\mathbf{x}) \;\approx\; (-3.2, -3.2, -3.2).$$

The estimator is far from $\mathbf{x}$ and pointing in the *opposite* direction. No statistician would actually report this estimate. Yet it has lower risk than δ^0 on average.

The positive-part estimator

Define

$$\delta^{JS+}(\mathbf{x}) \;=\; \left(1 - \frac{p-2}{\|\mathbf{x}\|^2}\right)^{+} \mathbf{x},$$

where $(a)^+ = \max(a, 0)$. The shrinkage factor is now clipped at zero: if shrinking $\mathbf{x}$ all the way to zero is closer than $\mathbf{x}$ itself, set $\delta^{JS+}(\mathbf{x}) = \mathbf{0}$.

Theorem (Baranchik 1964; Berger $§$5.4.3)

For $p \geq 3$, δ^{JS+} dominates δ^{JS}:

$$R(\theta, \delta^{JS+}) \;\leq\; R(\theta, \delta^{JS}) \quad \text{for all } \theta \in \mathbb{R}^p,$$

with strict inequality at some θ (specifically wherever $\mathbb{P}_\theta(\|\mathbf{X}\|^2 < p-2) > 0$).

Idea of proof. On the event $\|\mathbf{X}\|^2 \geq p-2$, the two estimators coincide. On the event $\|\mathbf{X}\|^2 < p-2$:

$$\|\delta^{JS+} - \theta\|^2 = \|\theta\|^2, \qquad \|\delta^{JS} - \theta\|^2 = \|(1-(p-2)/\|\mathbf{X}\|^2)\mathbf{X} - \theta\|^2$$

Three nested estimators

- δ^0: MLE; inadmissible for $p \geq 3$.
- δ^{JS}: James-Stein; dominates δ^0 but inadmissible.
- δ^{JS+}: positive-part Stein; dominates δ^{JS} but still inadmissible.

Is δ^{JS+} admissible? **No.** It can be improved further (proof by a similar SURE argument with a smooth substitute for the indicator).

In fact, **no estimator in the multivariate normal mean problem is uniformly admissible AND minimax** with a simple closed form. The best one can do is admissible *generalized Bayes* rules with specific priors (Strawderman 1971; Berger $§$8.9.2).

Stein's phenomenon in statistical practice

What broke

Before Stein (1956), it was assumed that:

- For each parameter component θ_i, the data component X_i gives a sufficient summary.
- Estimating each θ_i separately by the "obvious" estimator (the sample mean of that coordinate) is optimal.
- High-dimensional problems are simply p instances of one-dimensional problems.

Stein's phenomenon showed all three of these to be **wrong**. Three or more nominally unrelated estimation problems, processed jointly, can be solved better than three separate one-dimensional problems.

The mechanism is *not* that the parameters are related. The phenomenon holds even if $\theta_1, \theta_2, \theta_3$ are conceptually unrelated (e.g., a soccer player's batting average in baseball, a city's electricity consumption, and the gravitational constant). The improvement is purely statistical: high-dimensional geometry creates structure where there appears to be none.

What Stein's phenomenon does and does not say

It does say: the sample mean is uniformly improvable in $p \geq 3$ dimensions under squared-error loss. The improvement is universal in θ.

It does not say: the sample mean is bad in absolute terms. The improvement is concentrated near a chosen shrinkage point; far from it, the gain is negligible.

It does not say: shrinkage is always reasonable in applications. A practitioner who has good reason to believe each θ_i is far from zero will see δ^{JS} shrink in the wrong direction and gain little.

It does say: if you process many parameters by the same default rule (automated procedures, A/B test pipelines, scientific surveys), you can pay a price. This is the long-run argument from Lecture 5 ($§$4.8.3) made concrete.

Modern echoes

Stein's phenomenon, half a century later, is the foundational insight behind many high-dimensional methods:

Empirical Bayes (Efron and Morris 1973, 1975). Estimate the prior from the data, then apply Bayes machinery. The resulting estimators look like James-Stein with $p-2$ replaced by a data-driven quantity.

Hierarchical Bayes. Place a prior on the hyperparameters of a multivariate prior. Posterior means recover Stein-style shrinkage.

Ridge regression and the lasso. Shrinkage toward zero as a default, with the regularization parameter chosen by cross-validation or SURE.

Borrowing strength across subgroups (small-area estimation, multi-arm clinical trials, multi-task learning). Stein-type pooling reduces error for all groups simultaneously.

Adversarial robustness. The geometry that makes Stein's phenomenon work also makes high-dimensional models vulnerable to small adversarial perturbations.

Summary

What we built across Lectures 9 and 10

Lecture 9.

- Stein's Identity: $\mathbb{E}_\theta[(\mathbf{X}-\theta)^T\mathbf{g}(\mathbf{X})] = \mathbb{E}_\theta[\nabla \cdot \mathbf{g}(\mathbf{X})]$, proved via integration by parts.
- SURE: $\hat{R}(\mathbf{x}) = p + 2\nabla \cdot \mathbf{g}(\mathbf{x}) + \|\mathbf{g}(\mathbf{x})\|^2$ unbiasedly estimates the risk for $\delta(\mathbf{x}) = \mathbf{x} + \mathbf{g}(\mathbf{x})$.
- Specialized to δ^{JS}: $\hat{R}^{JS}(\mathbf{x}) = p - (p-2)^2/\|\mathbf{x}\|^2$.

Lecture 10.

- $\|\mathbf{X}\|^2$ is noncentral $\chi_p^2(\|\theta\|^2)$; admits Poisson mixture of central chi-squareds.
- $R(\theta, \delta^{JS}) = p - (p-2)^2\mathbb{E}_\theta[1/\|\mathbf{X}\|^2] < p$ for all $\theta \in \mathbb{R}^p$ when $p \geq 3$.
- The optimal shrinkage factor in $(1 - c/\|\mathbf{x}\|^2)\mathbf{x}$ is $c = p - 2$.
- Positive-part variant δ^{JS+} dominates δ^{JS} further.
- Geometric explanation: high-dimensional thin-shell concentration.
- Real-world implications: empirical Bayes, ridge, lasso, borrowing strength.

Reading

Berger, Section 5.4.3.

Foundational papers: Stein (1956) "Inadmissibility of the usual estimator..." (*Proc. Third Berkeley Symp.*); James and Stein (1961) "Estimation with quadratic loss" (*Proc. Fourth Berkeley Symp.*); Stein (1981) "Estimation of the mean of a multivariate normal distribution" (*Annals of Statistics*).

Modern perspective: Efron and Morris (1977) "Stein's paradox in statistics" (*Scientific American*) gives the classic accessible exposition. Efron (2010) *Large-Scale Inference* puts Stein's phenomenon at the foundation of modern statistical practice.

Exercises: Berger $§$5.4 Exercises 54, 55, 56, 57.